

Star Cluster Evolution – Revealed By GAIA

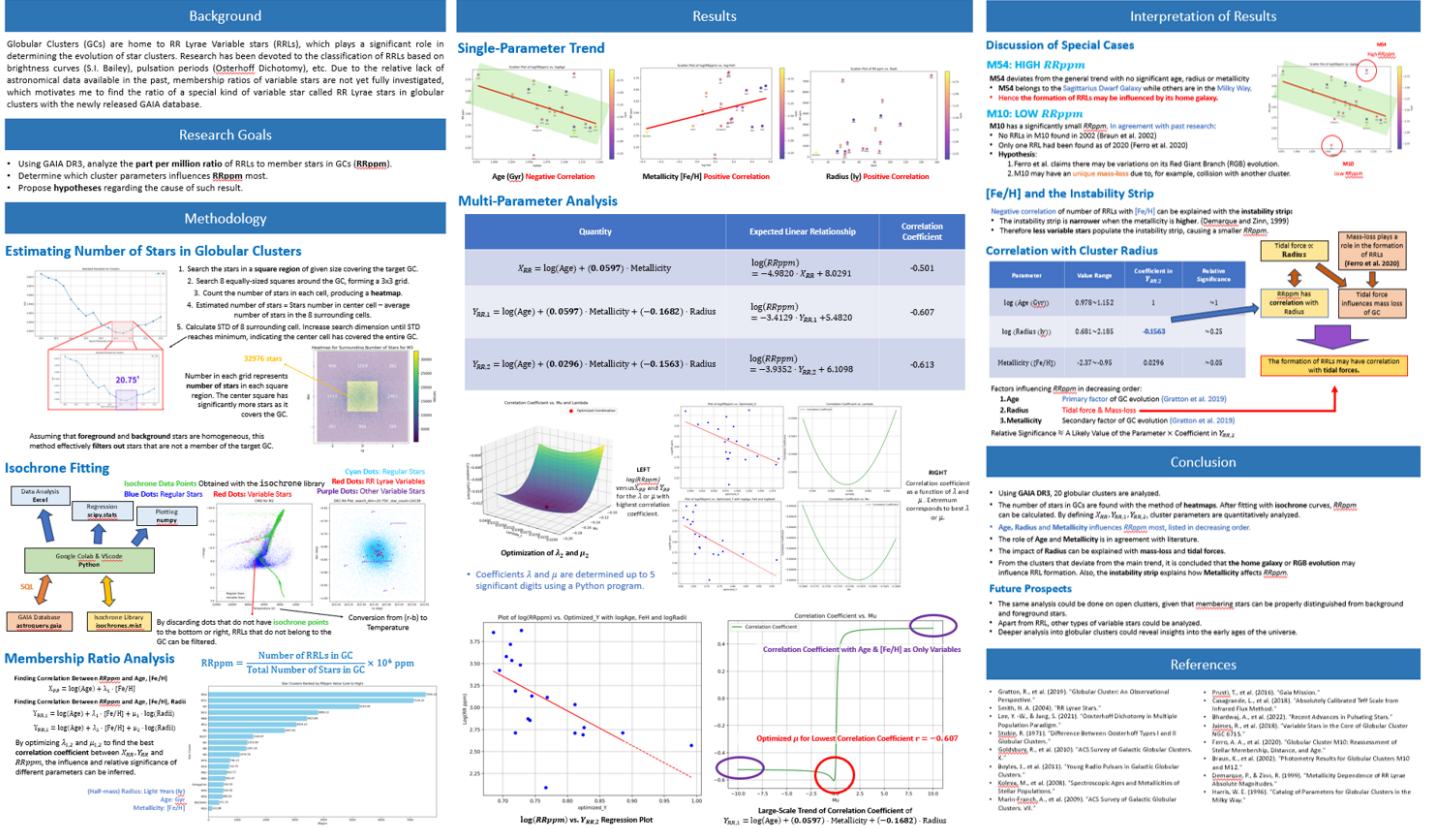
Project Data Book

Cheng-You Ho, Taiwan

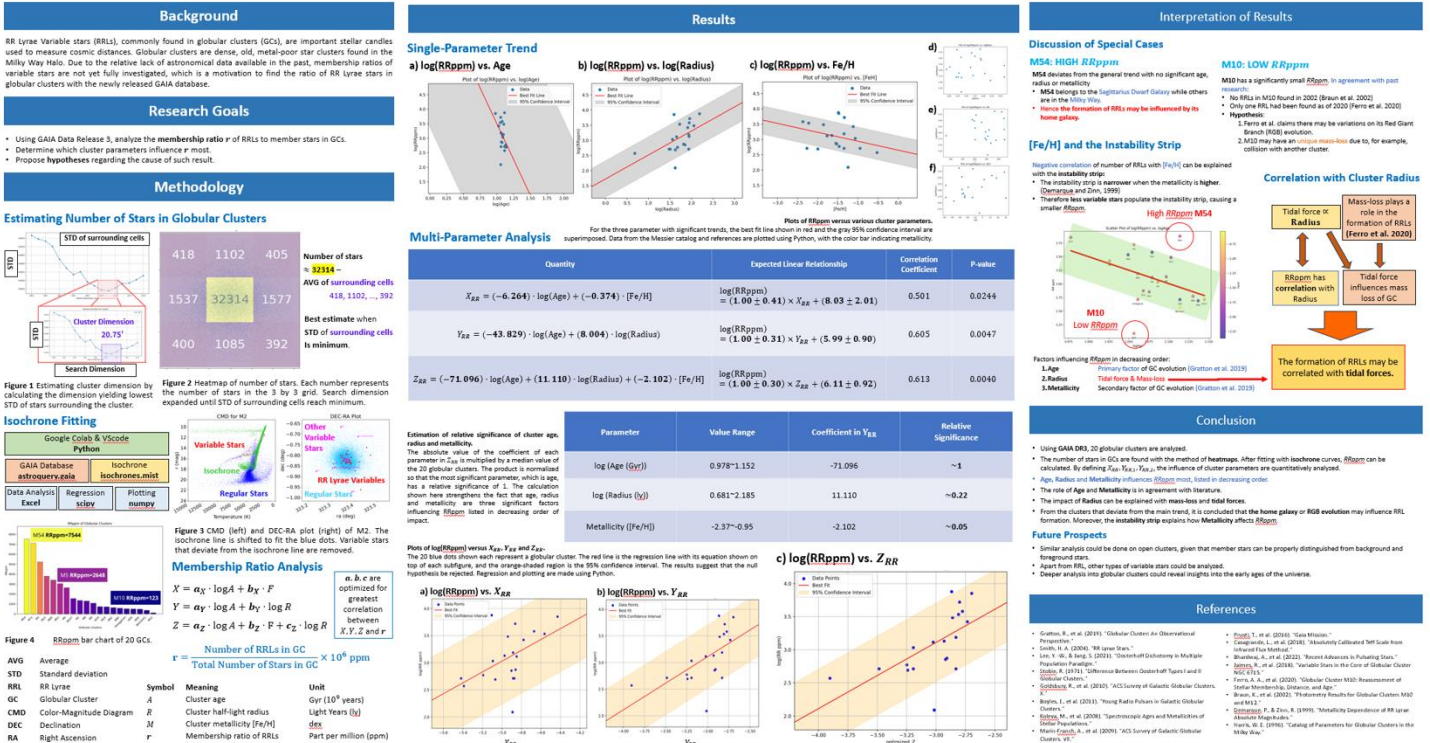
MOSTRATEC 2024

Poster Versions

January 2024 (TISF format)



September 10th, 2024 (1st revision with Professor Li, in MOSTRATEC format)



Background

RR Lyrae variable stars (RRLs), commonly found in globular clusters (GCs), are important stellar candles used for measuring cosmic distances. Globular clusters, which are dense, old, and metal-poor star clusters, are primarily located in the Milky Way's halo. Due to the limited astronomical data available in the past, the membership ratios of variable stars have not been fully investigated. This motivates the search for the ratio of RRL stars in globular clusters using the newly released GAIA database.

Research Goals

- Using GAIA Data Release 3, analyze the membership ratio R of RRLs in GCs.
- Determine which cluster parameters influence R most.
- Propose hypotheses regarding the cause of such result.

Methodology

Estimating Number of Stars in Globular Clusters

Membership Ratio Analysis

Figure 3 CMD (left) and DEC-RA plot (right) of cluster M2. The isochrone line was shifted to fit the blue dots. Variable stars that deviate from the isochrone line were removed.

Figure 4 Bar chart of R for 20 GCs.

Key Physical Parameters

Physical Parameter	Symbol	Unit
Age	A	Gyr (10^9 years)
Metallicity [Fe/H]	M	dex
Half-light radius	r	Light Years (ly)
Membership ratio of RRLs	R	Parts per million (ppm)

Figure 4 Bar chart of R for 20 GCs.

Results

Figure 5

Plots of (a) $\log(R)$ vs. $\log(A)$, (b) $\log(R)$ vs. $\log(R)$, and (c) $\log(R)$ vs. F . Each blue dot represents a globular cluster. The red line is the linear regression, and the yellow shaded region is the 95% confidence interval. Age and metallicity are negatively correlated to the membership ratio, while radius is positively correlated to it.

Figure 6

Plots of (a) $\log(R)$ vs. X , (b) $\log(R)$ vs. Y , and (c) $\log(R)$ vs. Z . Each blue dot represents a globular cluster. The red line is the linear regression, and the yellow shaded region is the 95% confidence interval. The fit suggests that $\log(R)$ is moderately correlated with X , Y and Z .

Table 1

Quantity	a	b	c	Correlation Coefficient	P-value
X	-6.264	-0.374		0.501	0.0244
Y	-43.829	8.004		0.605	0.0047
Z	-71.096	11.110	-2.102	0.613	0.0040

Table 2

Parameter	Value Range	Coefficient in Z	Relative Significance
$\log(A)$	0.978~1.152	-71.096	~ 1
$\log(R)$	0.681~2.185	11.110	~ 0.22
F	-2.37~0.95	-2.102	~ 0.05

Interpretation

High R : M54 Cluster

M54 has a high membership ratio with no significant age, radius or metallicity.

- M54 belongs to the Sagittarius Dwarf Galaxy while others are in the Milky Way.
- The formation of RRLs may be influenced by its home galaxy.

Low R : M10 Cluster

M10 has a significantly small R . In agreement with past research:

- No RRLs in M10 found in 2002 [1].
- Only one RRL found as of 2020 [2].
- Hypothesis: 1.R is claimed [2] that there may be variations on its Red Giant Branch (RGB) evolution. 2.M10 may have a unique mass loss due to, for example, collision with another cluster.

[Fe/H] and the Instability Strip

Negative correlation of membership ratio of RRLs with [Fe/H] can be explained with the instability strip:

- The instability strip is narrower when the metallicity is higher. [1]
- Therefore less variable stars populate the instability strip, causing smaller R .

Cluster Parameters Influencing R

- Age
- Radius
- Metallicity

Primary factor of GC evolution [4]
Secondary factor of GC evolution [4]

Conclusions

The formation of RRLs may be correlated with tidal forces.

Background

RR Lyrae variable stars (RRLs), commonly found in globular clusters (GCs), are important stellar candles used for measuring cosmic distances. Globular clusters, which are dense, old, and metal-poor star clusters, are primarily located in the Milky Way's halo. Due to the limited astronomical data available in the past, the membership ratios of variable stars have not been fully investigated. This motivates the search for the ratio of RRL stars in globular clusters using the newly released GAIA database.

Goals

- Using GAIA Data Release 3, analyze the membership ratio R of RRLs in GCs.
- Determine which cluster parameters influence R most.
- Propose hypotheses regarding the cause of such result.

Methodology

Estimating Number of Stars in Globular Clusters

Membership Ratio Analysis

Figure 3 CMD (left) and DEC-RA plot (right) of cluster M2. The isochrone line was shifted to fit the blue dots. Variable stars that deviate from the isochrone line were removed.

Figure 4 Bar chart of R for 20 GCs.

Key Physical Parameters

Physical Parameter	Symbol	Unit
Age	A	Gyr (10^9 years)
Metallicity [Fe/H]	M	dex
Half-light radius	r	Light Years (ly)
Membership ratio of RRLs	R	Parts per million (ppm)

Figure 4 Bar chart of R for 20 GCs.

Star Cluster Evolution — Revealed by GAIA

Cheng-Yu Ho, Taiwan

Results

Figure 5

Plots of (a) $\log R$ vs. X and (b) $\log R$ vs. Z . Each blue dot represents a globular cluster. The slope is normalized to 1. Most clusters fall inside the 95% confidence interval. Red line: Regression fit. Yellow shaded region: 95% confidence interval.

Figure 6

Plots of (a) $\log R$ vs. $\log A$, (b) $\log R$ vs. $\log R$, and (c) $\log R$ vs. M . Each blue dot represents a globular cluster. Age and metallicity are negatively correlated to the membership ratio, while radius is positively correlated to it.

Table 1

Quantity	a	b	c	Correlation Coefficient
X	-6.26	-0.37	-	0.501
Z	-71.11	-0.21	1.11	0.613

Table 2

Parameter	Value Range	Median Value	Coefficient in Z	Relative Significance
$\log A$	0.978~1.152	1.08	-71.11	~ 1
$\log R$	0.681~2.185	1.77	1.11	~ 0.26
M	-2.37~0.95	-1.51	-0.21	~ 0.04

Interpretations

High R : M54 Cluster

M54 has a high R . M54 belongs to the Sagittarius Dwarf Galaxy while others are in the Milky Way.

- The formation of RRLs may be influenced by its home galaxy.

Low R : M10 Cluster

M10 has a significantly small R . In agreement with past research [1][2]:

- There may be variations on its Red Giant Branch evolution [2].
- M10 may have a unique mass loss due to collision with another cluster.

[Fe/H] and the Instability Strip

Negative correlation of membership ratio of RRLs with [Fe/H] can be explained with the instability strip:

- The instability strip is narrower when the metallicity is higher. [1].
- Therefore less variable stars populate the instability strip, causing smaller R .

Conclusions

The formation of RRLs may be correlated with tidal forces.

Star Cluster Evolution

— Revealed by **GAIA**

Cheng-You Ho, Taiwan

RR Lyrae variable stars (RRLs), commonly found in globular clusters (GCs), are important stellar candles used for measuring cosmic distances. Globular clusters, which are dense, old, and metal-poor star clusters, are primarily located in the Milky Way's halo. Due to the limited astronomical data available in the past, the membership ratios of variable stars have not been fully investigated. This motivates the search for the ratio of RR Lyrae stars in globular clusters using the newly released GAIA database.

- Using GAIA Data Release 3, analyze the **membership ratio** R of RRLs in GCs
- Determine which cluster parameters influence R most.
- Propose **explanations** regarding the cause of such result.

Estimating Number of Stars in Globular Clusters

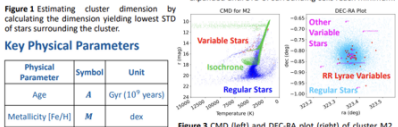


Figure 2 consists of three scatter plots labeled (a), (b), and (c). Each plot shows the relationship between $\log R$ (y-axis) and a different variable (x-axis). A red line represents the regression, and a yellow shaded region represents the 95% confidence interval.

- (a)** $\log R$ vs $\log A$: The x-axis ranges from 0.0 to 3.0, and the y-axis ranges from 1.5 to 4.0. The regression line shows a negative correlation.
- (b)** $\log R$ vs M : The x-axis ranges from -2.5 to -0.5, and the y-axis ranges from 2.0 to 4.0. The regression line shows a negative correlation.
- (c)** $\log R$ vs $\log r$: The x-axis ranges from 0.5 to 3.0, and the y-axis ranges from 2.0 to 4.0. The regression line shows a positive correlation.

Figure 5 Plots of (a) $\log R$ vs. $\log x$, (b) $\log R$ vs. M , and (c) $\log R$ vs. $\log r$. Each blue dot represents a globular cluster. Age and metallicity are negatively correlated to the membership ratio, while radius is positively correlated to it.

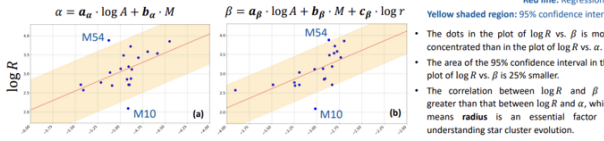


Figure 6 Plots of (a) $\log R$ vs. α and (b) $\log R$ vs. β . Each blue dot represents a globular cluster. The slope is normalized to 1. Most clusters fall inside the 95% confidence interval. * Clusters M54 and M10 have a high and low α and β respectively.

Table 1 Optimal coefficients and correlation coefficient between α , β and $\log R$.				
Quantity	α	β	c	Correlation Coefficient
α	-6.26	-0.37	-	0.501
β	-7.11	-0.21	1.11	0.613

High R: M54 Cluster

Red dots: RRLs

M54 has a high R:

- M54 belongs to the **Sagittarius Dwarf Galaxy** while others are in the **Milky Way**
- The formation of RRLs may be influenced by its home galaxy.

Low R: M10 Cluster

Red dots: RRLs

M10 has a significantly small R. In agreement with past research [12]:

- There may be variations on its **Red Giant Branch** position [2].
- M10 may have an **unique mass loss** due to collision with another cluster.

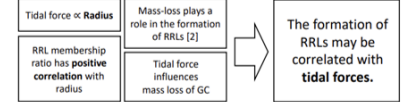
Negative correlation of membership ratio of RRLs with $[\text{Fe}/\text{H}]$ can be explained with the **instability strip**:

- The instability strip is **narrower** when the metallicity is **higher** [3].
- Therefore **less variable stars** populate the instability strip, causing a smaller R .

Table 3 Parameters and weights of previous result and this research.

	Previous Result [4]			This Work		
Parameter	A	M	r	A	M	r
Weight	$\equiv 1$	2	NA	$\equiv 1$	3	2

NA: Not Applicable *A*: Age *M*: Metallicity *r*: Half-Light Radius



- 20 globular clusters were analyzed using GAIA Data Release 3.
- I found that **radius** is a key factor that must be considered in understanding star cluster evolution. The impact of radius is mainly from the **mass-loss** and **tidal forces**.
- The **instability strip** explains how **metallicity** affects the RR Lyrae membership ratio.
- In addition to age and metallicity, I found that **radius** is even more essential than metallicity in analyzing the membership ratio of RR Lyrae stars.

- Similar analyses could be conducted on open clusters, given that member stars can be properly distinguished from background and foreground stars.
- Applying this method to consider the effect from radius to other star clusters.

- [1] K. Braun, M. Matsu K. Chiboucas, A. Athey, and D. Hurley-Keller, “Photometry Results for the Globular Clusters M10 and M12: Extinction Maps, Color-Magnitude Diagrams, and Variable Star Candidates,” *Astron J.*, 124(4), 2067–2082, 2002.
- [2] A. Ferro *et al.*, “The globular cluster M10: reassessment of stellar membership, distance, and age using its variable and HB stars,” *Mon Not R Astron Soc*, 499(3), 4026–4039, 2020.
- [3] P. Demarque, R. Zinn, Y.-W. Lee, and S. Yi, “The Metallicity Dependence of RR Lyrae Absolute Magnitudes from Synthetic Horizontal-Branch Models,” *Astron J.*, 119(3), 1398, 2000.
- [4] R. Gratton, A. Bragaglia, E. Carretta, V. D’Orazi, S. Lucatello, and A. Solima, “What is a globular cluster? An observational perspective,” *Astron Astrophys Rev*, 27(1), 1–336, 2019.

This work has made use of data from the European Space Agency (ESA) mission Gaia (<https://www.cosmos.esa.int/gaia>). Processed by the Gaia Data Processing and Analysis Consortium (DPAC, <https://www.cosmos.esa.int/web/gaia/dpac/consortium>). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement.

我與老師大部分的討論使用 Facebook Messenger。對話的圖片是 Facebook 提供的對話紀錄 HTML 檔案截圖。帳戶名稱已在 HTML 檔中修改為「學生」(我)與「老師」，以確保匿名。

Notes:

- The teacher agrees that the conversation snapshots be presented here.
- In the research paper, variables α and β were discussed. I originally named α as X_{RR} , so X_{RR} in this document represents α . The variable β was previously named Z_{RR} .

日期 Date	討論內容與進展 Discussion and Progress	圖片 Snapshots
2023/07/06~ 2023/08/05	我和老師花了一個月的時間討論與決定研究主題。 老師建議的主題有恆星、宇宙學與星團。經過討論，決定研究星團內的恆星。由於我有 Python 與 MySQL 的經驗，我和老師決定分析資料庫。 My teacher and I spent a month discussing and deciding on the research topic. Topics suggested by teachers include internal mechanisms inside stars, cosmology and star clusters. After discussion, it was decided to study the stars within the cluster. Since I have experience with Python and MySQL, my teacher and I decided to analyze the database.	老師 原理不難 7月06, 2023 10:59:24下午 老師 我可以查一下 7月06, 2023 10:59:20下午 老師 但這不是宇宙學 7月06, 2023 10:59:16下午 學生 是的，但他們好像只有分析資料庫而已 7月06, 2023 10:58:47下午 老師 系外行星？ 7月06, 2023 10:58:02下午 學生 https://www.snexplores.org/article/2023-regeneron-isef-winners-exoplanets-mpox-insect-pests-suicide-prevention 這是最近做宇宙學得獎的題目，不知道老師有沒有類似的主題？ Six young researchers took home the top awards, each valued at a minimum of \$50,000. Hundreds more shared nearly \$9 million in prizes at international event. https://www.snexplores.org/article/2023-regeneron-isef-winners-exoplanets-mpox-insect-pests-suicide-prevention 7月06, 2023 6:32:00下午 學生 像是有找到這個： https://esahubble.org/images/potw2321a/ A jellyfish galaxy adrift https://esahubble.org/images/potw2321a/ 8月06, 2023 12:26:15下午 學生 還有移動速率等參數 8月06, 2023 12:25:53下午 學生 我想分析一些較新發現的星團的組成 8月06, 2023 12:25:45下午 老師 那想做星團的什麼？ 8月06, 2023 11:57:46上午 學生 另外我有 MySQL 和 Python 的基礎因此操作資料庫我應該可以 8月05, 2023 2:35:22下午 學生 老師好，我把我能找到的和星團有關的科展都看過了 8月05, 2023 2:34:51下午

2023/08/06	瀏覽近期相關的科展作品，我們最終決定使用 GAIA DR3 資料庫分析球狀星團中的星體。 參考相關論文，我也草擬了一個大致的研究題目： Revisiting the Variable Stars in star clusters with GAIA data。 確認方向後，我開始上網找尋有關星團、變星等資料，充實基本知識。 Browsing recent related science exhibition works, we finally decided to use the GAIA DR3 database to analyze stars in globular clusters. Referring to relevant papers, I also drafted a general research topic: Revisiting the Variable Stars in star clusters with GAIA data. After confirming the direction, I started searching online for information about star clusters, variable stars, etc. to enrich my basic knowledge.	老師 這是gaia的data release 8 月 06, 2023 8:55:08下午 老師 https://www.cosmos.esa.int/web/gaia/dr3 Gaia Data Release 3 contents summary https://www.cosmos.esa.int/web/gaia/dr3 8 月 06, 2023 8:55:03下午 老師 因為新 8 月 06, 2023 8:53:29下午 老師 做gaia 是一個不錯的idea 👍 學生 8 月 06, 2023 8:53:26下午 老師 https://en.wikipedia.org/wiki/Gaia_(spacecraft) Gaia is a space observatory of the European Space Agency (ESA), launched in 2013 and expected to operate until 2025. The spacecraft is designed for astrometry: measuring the positions, distances and motions of stars with unprecedented precision,[5][6] and the positions of exoplanets by measuring att... https://en.wikipedia.org/wiki/Gaia_(spacecraft) 8 月 06, 2023 8:52:57下午 老師 GAIA 是相對較新的資料庫 8 月 06, 2023 8:52:37下午 老師 如 天琴座RR型變星 會出現在什麼年齡段的星團？ 8 月 06, 2023 9:39:11下午 老師 預期是 不同年齡的星團 的變星種類的關係 8 月 06, 2023 9:38:46下午 老師 我草擬一個題目 Revisiting the Variable Stars in star clusters with GAIA data 8 月 06, 2023 9:38:16下午 學生 好的! 8 月 06, 2023 9:36:40下午 老師 再延伸一些 8 月 06, 2023 9:35:44下午 老師 上面有一個type of variable star 先看一看 8 月 06, 2023 9:35:39下午 老師 先唸變星的分類與其演化的關係 8 月 06, 2023 9:35:19下午
------------	--	---

2023/08/07~
2023/08/08

上網搜尋教學網站後，我學會在 Google Colaboratory 中得到 GAIA DR3 的資料，包含個星體的 source_id、dec、ra、光偵測器數據等。

過程中，我們發現某些行星的自行量是負的，可能代表數據有缺陷。這使我們後來使用其他方式篩選成員星。

After searching online for tutorials, I learned to obtain GAIA DR3 data from Google Colaboratory, including the source_id, dec, ra, light detector data, etc. of individual stars.

During the process, we found that the proper motions of some planets are negative, which may indicate flaws in the data. This led us to later use other methods to filter member stars.

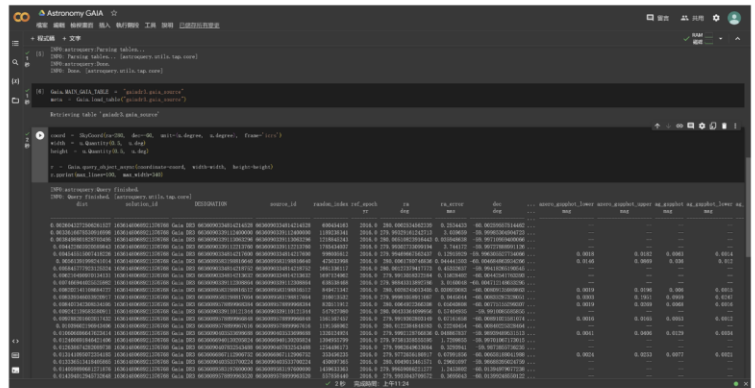
學生

不知道要怎麼得到年齡?
8月07, 2023 11:26:30上午

學生

但是好像只有位置的資訊
8月07, 2023 11:26:11上午

學生



8月07, 2023 11:25:26上午

學生

老師好，我有嘗試抓 GAIA 資料庫了(用 python)
8月07, 2023 11:25:23上午

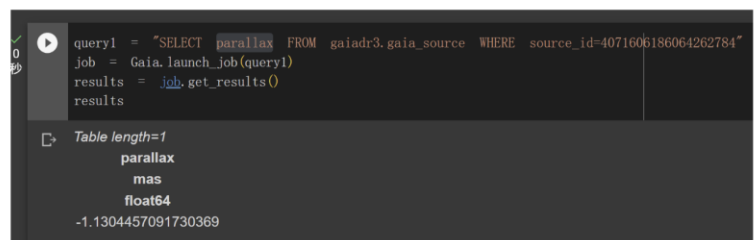
老師

這我真的不知道
8月08, 2023 8:55:38下午

老師

自行不是應該是絕對值嗎 QQ
8月08, 2023 8:55:24下午

學生



8月08, 2023 8:54:56下午

學生

老師我有遇到 parallax 是負值的星星，不知道正不正常?
8月08, 2023 8:54:55下午

老師

把v取代為r ok
8月08, 2023 7:50:08下午

2023/08/08

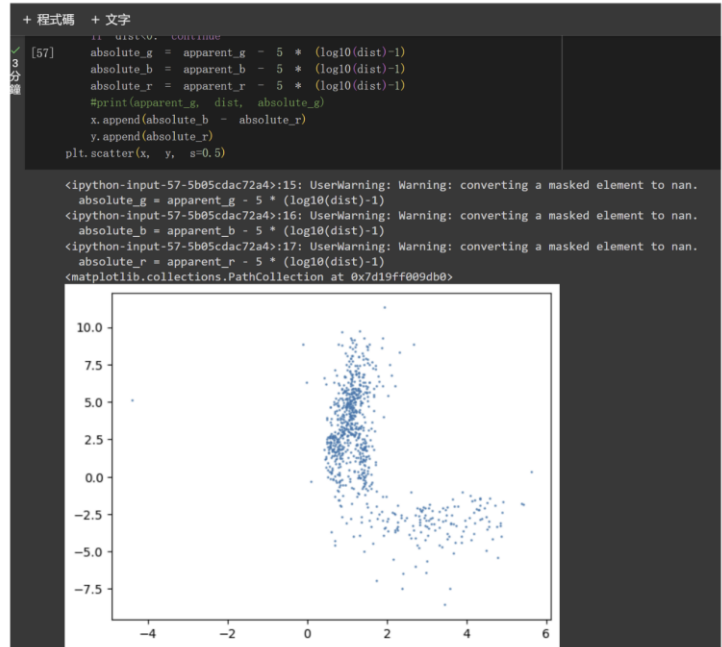
我根據資料庫的數據繪出初步的 CMD，並老師討論球狀星團 CMD 與 HRD 的圖形，對必備知識有更深的了解。

I drew a preliminary CMD based on the data in the database, and discussed with the teacher the CMD and HRD plots of globular clusters, which gave me a deeper understanding of the necessary knowledge.

學生

這是我嘗試畫的 v - (b-v) 圖
8 月 08, 2023 8:57:44 下午

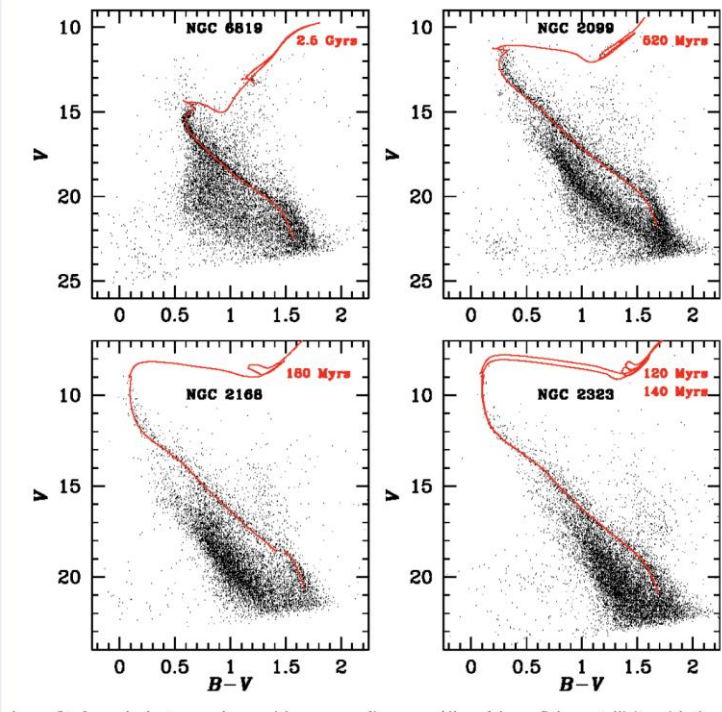
學生



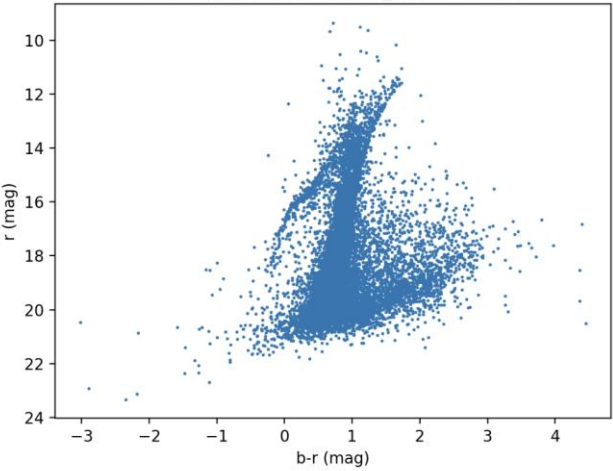
老師

他標了一條主序帶 對吧？
8 月 08, 2023 9:23:10 下午

老師



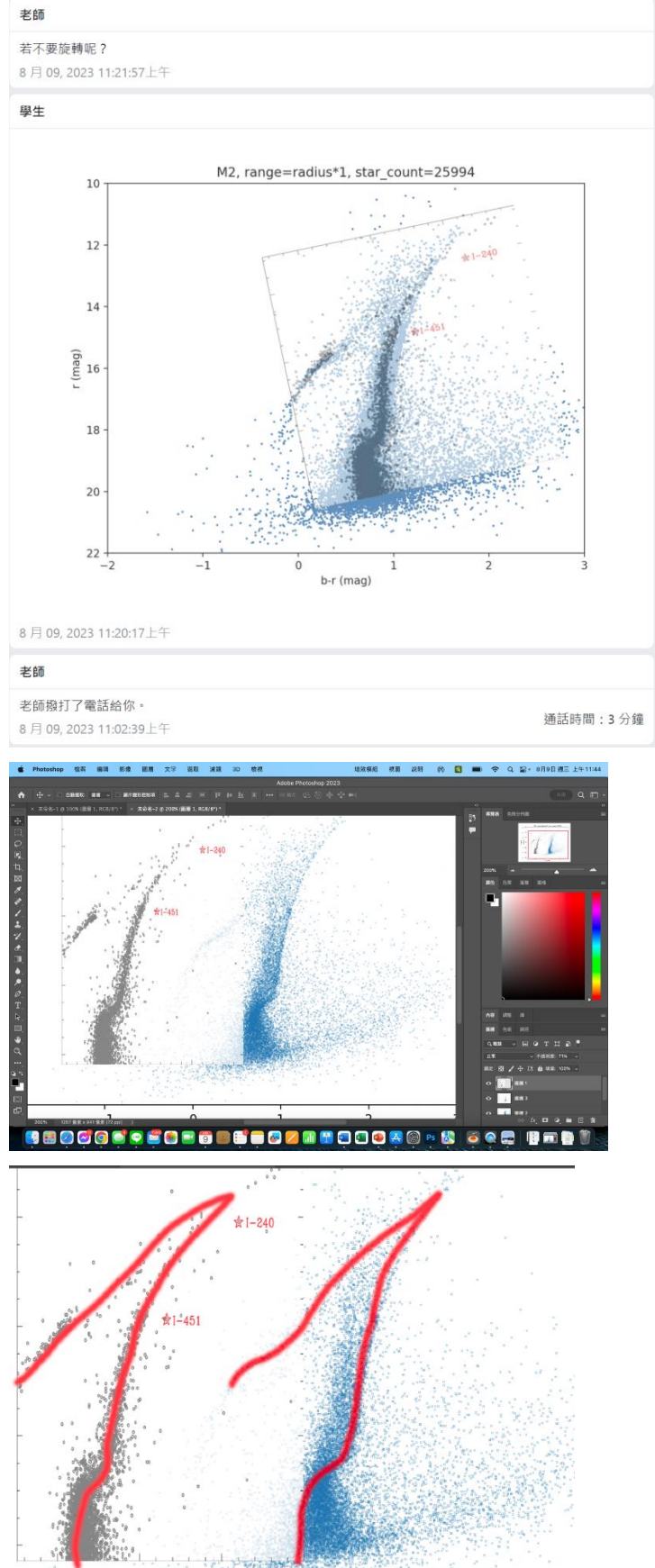
2023/08/08	意識到篩除非成員星或許是本研究中將來會遇到的一大難題。 Realizing that filtering out non-member stars may be a major problem that will be encountered in this study in the future.	老師 一般的星星是做不出HRD的 8 月 08, 2023 9:21:13 下午 學生 是指不能直接從CMD或HRD看出來嗎? 是不是還要另外分析其他東西 8 月 08, 2023 9:20:48 下午 老師 只能說有一些特徵，比如說自行、成份（必然相近）、與背景星的距離（同一星團距離相近） 8 月 08, 2023 9:18:52 下午 老師 這是一個難題 8 月 08, 2023 9:18:20 下午 老師 所以之前那些科展，是不是有一些就是在做成員星？ 8 月 08, 2023 9:18:16 下午 老師 你講到重點了！！ 8 月 08, 2023 9:18:04 下午 老師 https://sites.astro.caltech.edu/~george/ay1/lec_pdf/Ay1_Lec09.pdf https://sites.astro.caltech.edu/~george/ay1/lec_pdf/Ay1_Lec09.pdf 8 月 08, 2023 9:17:57 下午 學生 那要怎麼分辨一群星星是不是星團 8 月 08, 2023 9:17:24 下午 老師 是的！ 8 月 08, 2023 9:25:36 下午 學生 所以明顯的部分反而是背景星? 8 月 08, 2023 9:25:30 下午 老師 但是這只能粗略分 8 月 08, 2023 9:23:50 下午 老師 那就是背景星 8 月 08, 2023 9:23:42 下午 老師 左下是不是有個寬胖的線 8 月 08, 2023 9:23:39 下午 老師 這四個是疏散星團，星星比較少 8 月 08, 2023 9:23:28 下午 老師 就那一條紅色的線 8 月 08, 2023 9:23:18 下午
-------------------	--	--

2023/08/08~ 2023/08/09	和老師決定使用 Messier Catalog 的星體。 初步嘗試，搜尋 M2 的赤經、赤緯後，搜尋給定半徑範圍內的星體，成功繪出 CMD，但同時也發現背景星問題嚴重。 Me and my teacher decided to use the stars from the Messier Catalog. A preliminary attempt was made to search for the right ascension and declination of M2, then search for stars within a given radius, and successfully draw the CMD. However, it was also found that the background star problem was serious.	老師 所以大概至少要抓 他的半徑的2.5倍以上 8 月 08, 2023 10:01:13下午 老師 半徑會有亮度半徑 潮汐半徑 引力半徑等 8 月 08, 2023 10:00:55下午 老師 星團 尤其是球狀星團 8 月 08, 2023 10:00:44下午 老師 這個半徑其實不準哦！ 8 月 08, 2023 10:00:37下午 老師 裡面會有半徑 8 月 08, 2023 10:00:31下午 老師 你先查一下 messier catalog 8 月 08, 2023 10:00:22下午 學生 通常要限制多少呢 8 月 08, 2023 10:00:08下午 學生 好的! 8 月 08, 2023 9:59:45下午 老師 看起來正常了 8 月 09, 2023 10:54:36上午 學生 M2, range=radius*1, star_count=25994  8 月 09, 2023 10:54:28上午 學生 我試試看 8 月 09, 2023 10:53:29上午
-----------------------------------	--	--

2023/08/09

將現有論文資料與透過
GAIA 繪出的 CMD 疊圖，
確認資料的正確性。

Overlay the existing paper
data with the CMD diagram
drawn through GAIA to
confirm the accuracy of the
data.



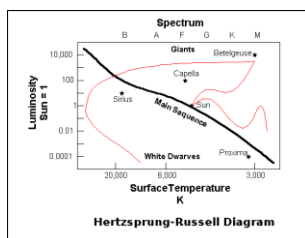
2023/08/09

調整繪圖參數與排版，繪出較漂亮的 CMD，並且發現清楚的 turn-off point。

比較相關資料，發現 GAIA 的資料確實比較充足。

Adjust the drawing parameters and layout to draw a more beautiful CMD and find a clear turn-off point.

Comparing relevant information, I found that GAIA's information is indeed sufficient.



老師

http://astronomy.nmsu.edu/jasonj/565/docs/11_07.pdf
%PDF-1.5 %
15 0 obj > stream x[KG_vDp@r#qJ#[]\$ztgUwJ+z/3k^YK?[]Q[VWl-Xo]_[6FqT
^%CynN6^3q3tN\$%Sg%Zwy:#7/ nK0hn)S^ba]ku3])Qoap6+d bQFg5x\gü
http://astronomy.nmsu.edu/jasonj/565/docs/11_07.pdf

8 月 09, 2023 9:16:07 下午

老師

對

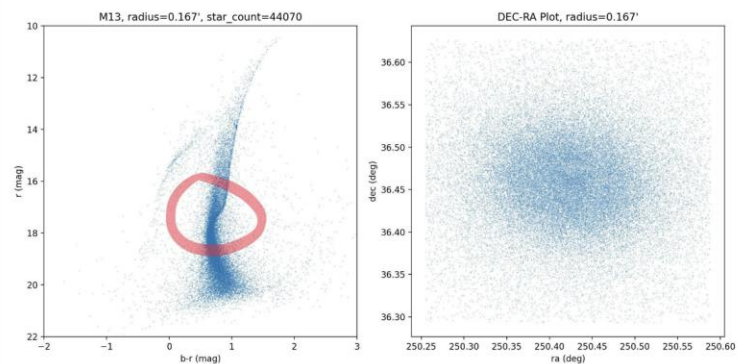
8 月 09, 2023 9:15:41 下午

學生

請問轉折是指這裡嗎？

8 月 09, 2023 9:15:22 下午

學生



8 月 09, 2023 9:15:13 下午

老師

比之前的類似的計畫都要厲害

8 月 09, 2023 9:26:40 下午

老師

但是你比較就會發現，gaia dr3 相當猛

8 月 09, 2023 9:26:30 下午

老師

不是“細細”的線，那就是背景

8 月 09, 2023 9:26:12 下午

老師

你細看一下最後這張圖的“轉折”把你的圖的橫軸放大（只看 -1~2）你會清楚的看到那一個變化

8 月 09, 2023 9:26:02 下午

老師

裡面有些以虛線標出來的地方，其實就是演化速度快，所以不是很清楚的地方

8 月 09, 2023 9:24:53 下午

老師

不同的位置有不同的意義

8 月 09, 2023 9:24:29 下午

老師

這個圖解析的比較清楚

8 月 09, 2023 9:24:23 下午

2023/08/10	Population I & II 與主序帶的討論。 初步尋找星團範圍內的變星，發現 GAIA 將變星分為 24 類。由於發現多數星團皆有天琴座 RR 型變星，我有意圖以此為主要的研究對象。 Discussion of Population I & II and Main Sequence. After a preliminary search for variable stars within the star cluster, it was found that GAIA classified variable stars into 24 categories. Since most star clusters are found to contain RR Lyrae variable stars, I intend to use this as the main research object.	老師 pop i 是多金屬 (佔總質量3%以上) 8 月 09, 2023 9:31:35下午 老師 pop ii 是少金屬 8 月 09, 2023 9:31:18下午 老師 是理論上無金屬的第一代恆星 8 月 09, 2023 9:31:09下午 老師 這一篇是做星族3 (polulation III) 8 月 09, 2023 9:31:00下午 老師 像這些就是從理論上 討論不同質量 不同金屬豐度的恆星演化 8 月 09, 2023 9:30:33下午 老師 https://www.researchgate.net/figure/Evolution-of-massive-Pop-III-stars-15-30-60-200-and-500-M-with-and-without_fig1_301312628 Download scientific diagram Evolution of massive Pop III stars (15, 30, 60, 200 and 500 M $\odot$) with and without rotation. The adopted initial rotational speeds in units of the Keplerian value are marked by different colors as indicated by the labels. These evolutionary models are taken from Yoon ... https://www.researchgate.net/figure/Evolution-of-massive-Pop-III-stars-15-30-60-200-and-500-M-with-and-without_fig1_301312628 8 月 09, 2023 9:30:05下午 老師 要去掉不合理的，所以改在cmd 中標出變星型 8 月 10, 2023 11:22:21上午 老師 有天琴座RR 8 月 10, 2023 11:21:57上午 老師 有造父 8 月 10, 2023 11:21:51上午 學生 BEST_CLASS_SCORE : Score of the best class (float) It describes a quantity between 0 and 1 which is related to the (median) normalised rank of the confidence of the classifier(s) in the identification of the best class (best_class_name). In the special case of class 'EP', all scores are set to 1. See Section 10.3.4.1 of the release documentation for details. 8 月 10, 2023 11:21:48上午 老師 score 是什麼？ 8 月 10, 2023 11:21:13上午 學生 這個就是 24 種分類 8 月 10, 2023 11:21:00上午 老師 有變星的類型嗎 8 月 10, 2023 11:20:47上午
-----------------------	---	---

2023/08/11

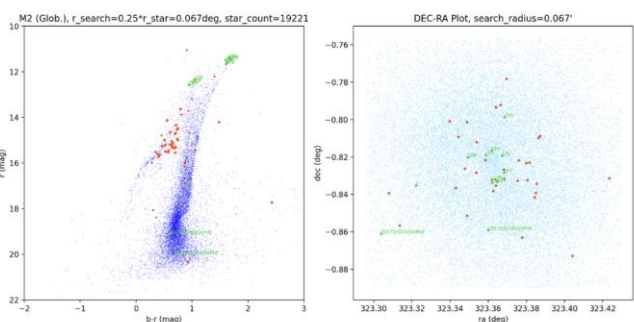
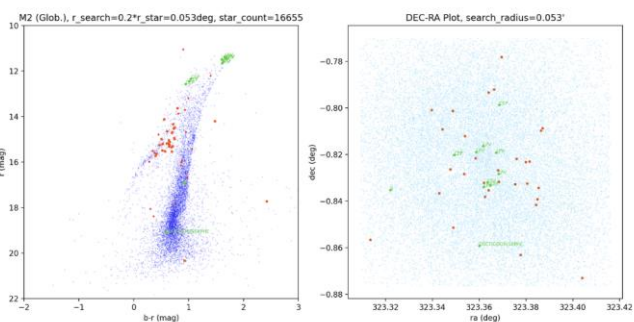
成功在 CMD 與 dec-ra 圖上標示各種類的變星。

透過分析 M44，我發現疏散星團較球狀星團難分析，因此決定以研究後者為主。

Successfully marked various types of variable stars on CMD and dec-ra diagrams.

Through the analysis of M44, I found that open star clusters are more difficult to analyze than globular star clusters, so I decided to focus on the latter.

學生



老師

1度

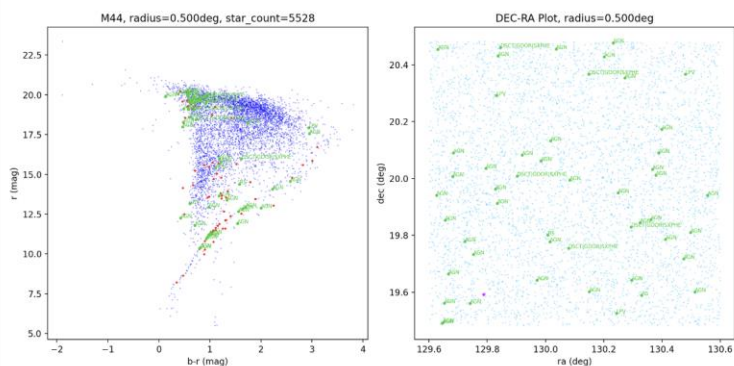
8月11, 2023 9:32:30下午

老師

m44 超大

8月11, 2023 9:32:27下午

學生



8月11, 2023 9:32:12下午

學生

老師這是我剛剛搜尋 M44(疏散)的結果，從右邊這張圖我看不太出來哪裡有星星聚集，左邊這張也和之前的很不一樣，請問這樣正常嗎？

8月11, 2023 9:32:12下午

2023/08/11~
2023/08/15

為篩選非成員星，老師首先建議以零齡主序帶 (zero age main sequence, ZAMS) 擬合，嘗試多種相關方程式後發現成效不彰。

我們認為原因可能是球狀星團年齡太老，不能使用 ZAMS 找出星體在 CMD 或 HRD 上應分布的位置。

In order to filter out non-member stars, my teacher first suggested fitting the zero age main sequence (ZAMS). After trying a variety of related equations, we found that the results were ineffective.

We believe that the reason may be that globular clusters are too old to use ZAMS to find out where the stars should be distributed on the CMD or HRD.

學生

老師好，這是直線分割的圖，我等等再試試看用 ZAMS 分割
8 月 12, 2023 5:43:08 下午

老師

那個 0 的意思是不考慮紅化消光
8 月 11, 2023 10:23:10 下午

老師

但原理是靠 (b-v)0
8 月 11, 2023 10:22:48 下午

老師

但沒有金屬
8 月 11, 2023 10:22:30 下午

老師

裡面有多種模式
8 月 11, 2023 10:22:26 下午

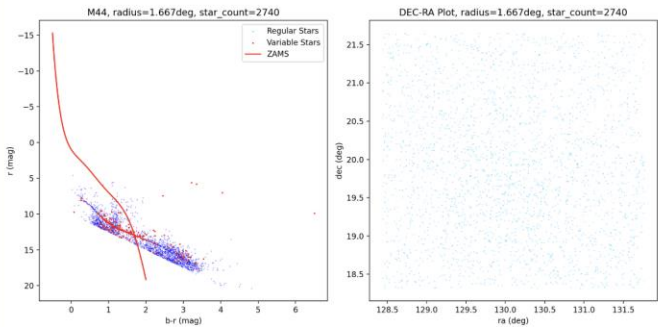
老師

要接受 cookie
8 月 11, 2023 10:21:03 下午

老師

https://xoomer.virgilio.it/hrtrace/Miscellanea1.htm#_top
當時應該是從這個網站找到方程式，用的是編號 0 "Schmidt-Kaler Luminosity Class V"
https://xoomer.virgilio.it/hrtrace/Miscellanea1.htm#_top
8 月 11, 2023 10:20:46 下午

學生

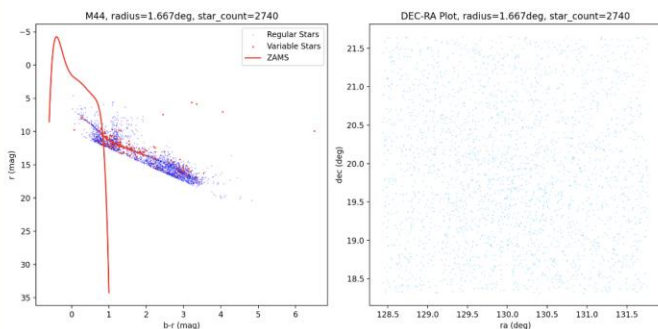


8 月 12, 2023 7:55:18 下午

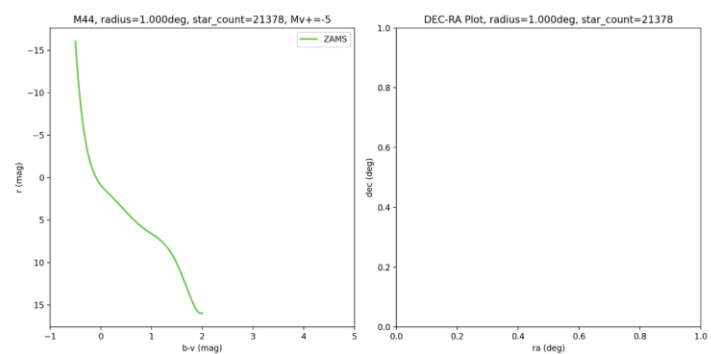
學生

這是 KEENAN 公式
請問這樣正常嗎?
8 月 12, 2023 7:55:18 下午

學生



學生



8月15, 2023 9:38:09上午

老師

你單畫一張zams出來

8月15, 2023 9:37:01上午

老師

我還是覺得怪

8月15, 2023 9:37:01上午

學生

這是用第八個公式的 ZAMS

8月15, 2023 9:33:44上午

老師

你這zams 不對

8月15, 2023 8:27:50上午

老師

不可以

8月15, 2023 8:27:44上午

學生

請問這樣可以嗎

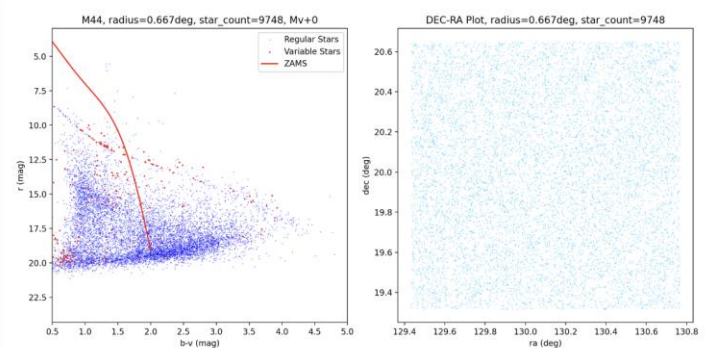
8月14, 2023 9:18:15下午

學生

但ZAMS和主序帶沒有很合耶

8月14, 2023 9:18:07下午

學生



2023/08/19~
2023/08/24

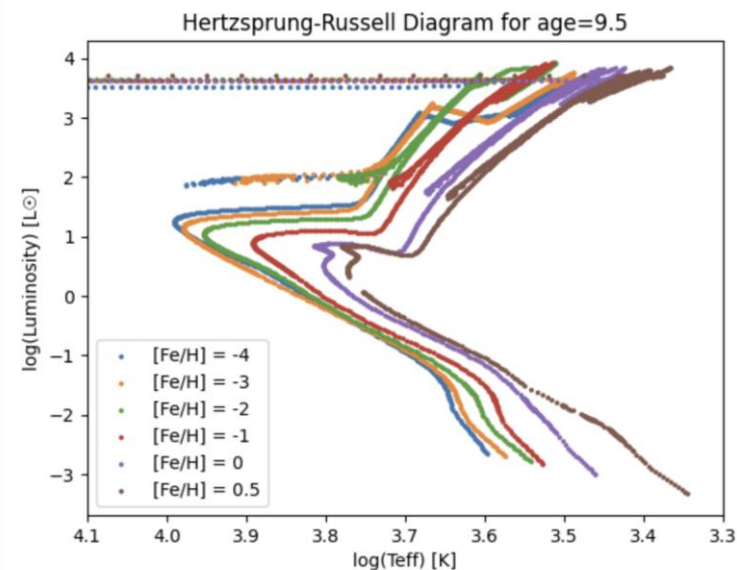
我從老師給的網站出發，找到了Python的 isochrone 模組。和老師討論後，我們認為 z 等年齡線 (isochrone) 比 ZAMS 更適合擬合球狀星團。

給定金屬豐度與年齡，isochrone 模組就能給出對應的等年齡線的點的集合。

I started from the website given by the teacher and found the isochrone module of Python. After discussing with the teacher, we believe that the z isochrone is more suitable for fitting globular star clusters than ZAMS.

Given metallicity and age, the isochrone module can give a set of points on the corresponding iso-age line.

學生



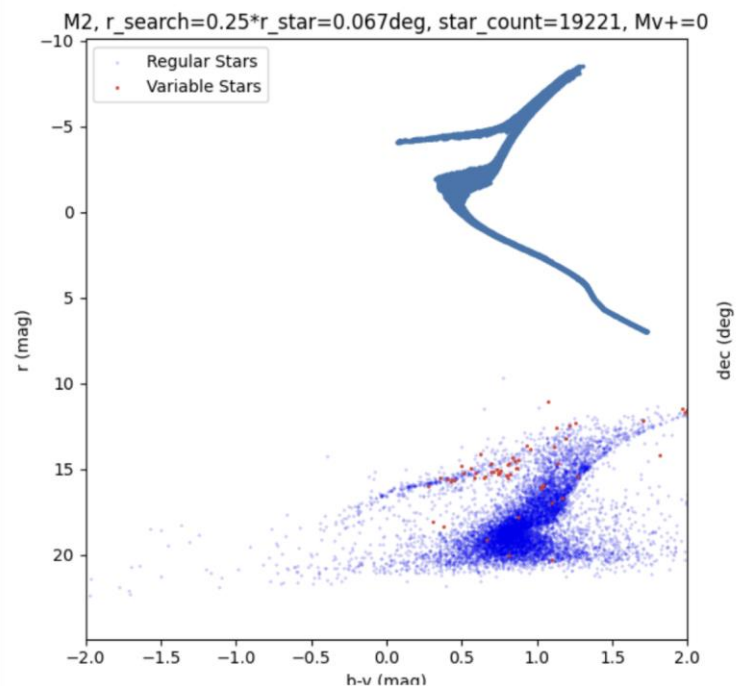
8月22, 2023 10:08:45下午

學生

上面是 isochrone 資料庫，下面是 gaia

8月24, 2023 6:03:42下午

學生



8月24, 2023 6:03:25下午

2023/08/24~
2023/08/28

由於都卜勒效應、紅化等因素，isochrone 與 GAIA 繪出的 CMD 必須平移後才能重合。我們一開始想用相關公式算出平移的幅度，再平移兩圖形，但最後發現這樣做困難重重，決定直接手動平移。

Due to Doppler effect, reddening and other factors, the CMD drawn by isochrone and GAIA must be translated before they can overlap. We initially wanted to use relevant formulas to calculate the magnitude of the translation, and then translate the two graphics, but in the end we found that it was difficult to do so and decided to translate directly manually.

學生

```
from dust_extinction.parameter_averages import F99
import numpy as np

# Define the wavelengths you're interested in (in microns)
wave = np.array([0.65]) # Example: red light

# Instantiate the extinction model and evaluate
ext = F99(Rv=3.1) # Rv=3.1 is a common value, but adjust as needed
a_lambda = ext(wave)
print(a_lambda)
```

[0.18417568]

8 月 26, 2023 9:48:12 上午

學生

老師好，我代 $R_v=3$ ，結果 0.65 1/micron 的紅光被修正到 0.18 1/micron，直接變成藍光，請問這樣正常嗎？

8 月 26, 2023 9:48:09 上午

學生

沒關係那我應該先用常數就好

8 月 25, 2023 9:04:10 下午

老師

最常見就是 AOV 型的星

8 月 25, 2023 9:01:54 下午

老師

你要找出標準星去校準

8 月 25, 2023 9:01:40 下午

老師

只能用函數下去猜

8 月 28, 2023 12:12:55 下午

老師

對，這裡的 AV 一定是錯的

8 月 28, 2023 12:12:49 下午

學生

那請問要怎麼用距離修正呢？

8 月 28, 2023 11:46:30 上午

學生

所以 AV 應該是錯的？

8 月 28, 2023 11:46:09 上午

老師

垂直方向為 距離加消光

8 月 28, 2023 7:30:47 上午

老師

沒有修正距離

8 月 28, 2023 7:30:35 上午

老師

等等！

8 月 28, 2023 7:30:28 上午

學生

我比較懷疑是不是 AV 的數值有錯？但如果 AV 確實是垂直的移動距離，應該是 20 左右沒錯

8 月 27, 2023 2:09:12 下午

2023/08/28

手動疊合後，我在老師的引導下，根據恆星在 CMD 上與 isochrone 資料點的距離與方向，設計了能大致篩選非成員星的條件 (rule)。

After manual overlay, under the guidance of the teacher, I designed a rule that can roughly filter out non-member stars based on the distance and direction of the star from the isochrone data point on the CMD.

老師

左下的那兩個點，要不要去掉？

8 月 29, 2023 11:42:58 上午

老師

給出一些rule

8 月 29, 2023 11:42:32 上午

老師

你要做的是去掉某一區的變星

8 月 29, 2023 11:42:24 上午

老師

感覺fitting還差了一點點

8 月 29, 2023 11:42:06 上午

老師

我的意思是，大概

8 月 29, 2023 11:40:50 上午

學生

是一大堆點

8 月 29, 2023 11:40:29 上午

學生

Isochrone 的線不是一個方程式

8 月 29, 2023 11:40:23 上午

老師

寫方程式如何？

8 月 29, 2023 11:40:09 上午

學生

是一大堆點

8 月 29, 2023 11:40:29 上午

學生

Isochrone 的線不是一個方程式

8 月 29, 2023 11:40:23 上午

老師

寫方程式如何？

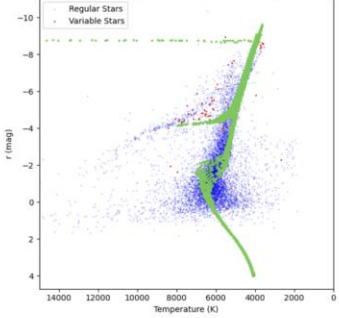
8 月 29, 2023 11:40:09 上午

學生

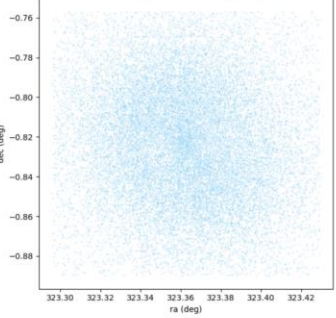
老師好，這是我做 fitting 的結果。目前剩下決定怎麼挑掉不要的點。因為x, y,軸單位不一樣，有一點困難

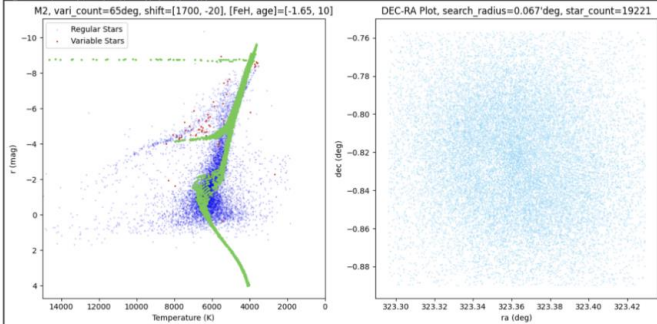
8 月 29, 2023 11:39:45 上午

M2, r_search=0.067deg, shift=[1700, -20], [FeH, age]=[-1.65, 10]



DEC-RA Plot, search_radius=0.067deg, star_count=19221



		老師 因為變星可能會變亮，所以是小於 0.5 8 月 29, 2023 11:44:59 上午 老師 或某個值 8 月 29, 2023 11:44:39 上午 老師 或0.5 8 月 29, 2023 11:44:32 上午 老師 就是離你的點大於1 就是不要 8 月 29, 2023 11:44:10 上午 老師 一個簡單的規則 8 月 29, 2023 11:43:54 上午 老師 左下兩個 右側兩個 8 月 29, 2023 11:43:40 上午 老師 有四個點 8 月 29, 2023 11:43:33 上午 老師 不在線附近或上方的 8 月 29, 2023 11:43:29 上午	
		學生 老師好，這是我新的程式碼 8 月 30, 2023 8:39:54 上午 學生 https://colab.research.google.com/drive/1undRWqeSxdT96mJXxE7NL_un4fE0ALpM?usp=sharing Colaboratory notebook https://colab.research.google.com/drive/1undRWqeSxdT96mJXxE7NL_un4fE0ALpM?usp=sharing 8 月 30, 2023 8:39:37 上午 學生 我再調調看好了 8 月 29, 2023 9:21:52 下午 老師 左上角要篩掉那麼多嗎？ 8 月 29, 2023 9:18:56 下午 老師 換一個星團呢？ 8 月 29, 2023 9:18:43 下午 學生 Discarded Variable Stars: 0 Return lengths: 65 65 /usr/local/lib/python3.10/dist-packages/numpy/memmap.py:2826: UserWarning: Warning: converting a masked element to nan. data = np.array(data, dtype=ndtype, copy=copy) M2, vari_count=65deg, shift=[1700, -20], [FeH, age]=[-1.65, 10] DEC-RA Plot, search_radius=0.067deg, star_count=19221 Regular Stars Variable Stars 	

2023/09/10~ 2023/09/20	大部分的程式完成後，我開始分析 Messier Catalog 中的 10 個星團。 此時我仍在嘗試分析球狀星團，過程中遇到了諸如 M103 等較不易分析的星團。 After most of the programming was completed, I started analyzing the 10 star clusters in the Messier Catalog. At this time I was still trying to analyze globular clusters, and in the process I encountered clusters such as M103 that were less easy to analyze.	學生 老師好，我目前的程式可以分析球狀星團了 但資料庫沒有疏散星團的 isochrone 請問我要怎麼篩選變星呢 9 月 11, 2023 9:10:00 上午 學生 或是老師有沒有印象課本中有沒有寫呢? 9 月 10, 2023 4:33:44 下午 學生 前人是否有做過 9 月 10, 2023 4:30:02 下午 學生 像是有沒有說球狀星團比較容易出現造父變星 等等的? 9 月 10, 2023 4:29:52 下午 老師 哪一顆? 9 月 10, 2023 4:29:10 下午 學生 老師哪請問對於球狀或疏散星團有什麼樣的星星 (或變星) 目前有沒有甚麼已知的理論呢? 9 月 10, 2023 4:21:14 下午 9 月 20, 2023 8:39:44 下午 學生 老師請問這個疏散星團的主序帶是哪一個呢 9 月 20, 2023 8:39:44 下午 老師 https://www.cysh.cy.edu.tw/p/404-1008-95376.php <p color="#000000" fontfamily="標楷體" fontsize="16" lineheight="150%" paddingtop="0" paragraphlevel="0" paragraphstartindent="48" rootdefinition="0" rowspacing="0" textindent="-48">主旨： 函轉國立臺灣科學教育館(以下簡稱科教館)辦理「2024年臺灣國際科學展覽會國內... https://www.cysh.cy.edu.tw/p/404-1008-95376.php 9 月 20, 2023 2:04:21 下午
--	---	--

2023/09/20~
2023/09/27

在 ChatGPT 3.5 協助資料整理下，我選定了 15 個 Messier Catalog 的球狀星團 (M2, M3, M9, M10, M19, M28, M30, M54, M56, M68, M72, M79, M80, M92, M107)，並將重點放在分析 RR 型天琴座變星 (RRL)。

此時，我開始將 CMD、dec-ra 圖、星團參數、等年齡線平移量等參數以 PowerPoint 彙整，以方便和老師討論。

不過，這個時候的程式碼尚有一些 bug。程式碼經過修正後，得到的數據才用在最終的分析上。

With the help of ChatGPT 3.5, I selected 15 Messier Catalog globular clusters (M2, M3, M9, M10, M19, M28, M30, M54, M56, M68, M72, M79, M80, M92, M107), and focus on the analysis of RR-type Lyrae variables (RRL).

At this time, I began to compile the CMD, dec-ra diagram, star cluster parameters, isoage line translation and other parameters in PowerPoint to facilitate discussion with the teacher.

However, the code at this time still has some bugs. After the program code was modified, the data obtained was used in the final analysis.

學生

目前注意到比較特別的就是 M3, M68 都有很多天琴座 RR 但兩個都不是天琴座的
然後這些 RR 分布很接近
9 月 23, 2023 6:02:00 下午

學生

下載檔案：
your_activity_across_facebook/messages/inbox/huangguanfu_10160632118289602/files/globularvari0923_296530459760790.pdf
9 月 23, 2023 6:01:38 下午

學生

下載檔案：
your_activity_across_facebook/messages/inbox/huangguanfu_10160632118289602/files/AgeComparisonGrand_7126152884061630.xlsx
9 月 23, 2023 6:01:31 下午

學生

老師好，這裡有十個球狀星團的資料
9 月 23, 2023 6:01:30 下午

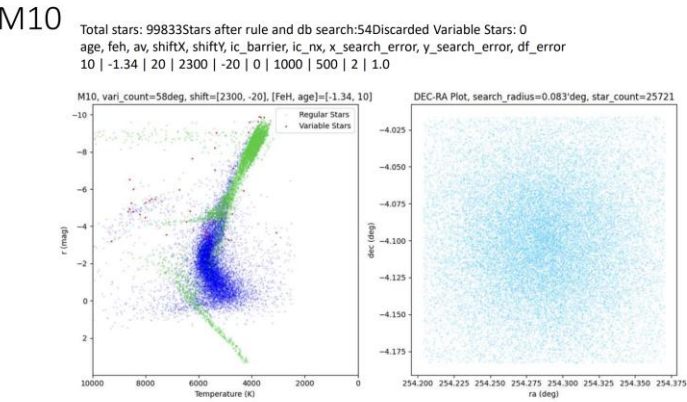
學生

好的
9 月 22, 2023 10:03:57 下午

Table of Globular Clusters

Name	ra_h	ra_m	ra_s	dec_deg	dec_min	dec_sec	dim_min	Age
M2	21	33	27.02	0	-49	-23.7	1612.5	Gyr
M3	13	42	11.62	28	22	38.2	1811.39	Gyr
M9	17	19	11.78	-18	-30	-58.5	9.3	
M10	16	57	8.92	-4	-5	-58.07	20	
M19	17	2	37.69	-26	-16	-4.6	17	
M28	18	24	32.89	-24	-52	-11.4	11.2	
M30	21	40	22.12	-23	-10	-47.5	12	
M54	18	55	3.33	-30	-28	-47.5	-1	
M56	19	16	35.57	30	11	5	-1	
M68	12	39	27.98	-26	-44	-38.6	-1	
M72	20	53	27.7	-12	-32	-14.3	1	
M79	5	24	10.59	-24	-31	-27.3	-1	
M80	16	17	2.41	-22	-58	-33.9	-1	
M92	17	17	7.39	43	8	9.4	-1	
M107	16	32	31.86	-13	-3	-13.6	-1	

M10



2023/09/20~ 2023/09/27	我發現大部分的星團除了有 RRL 外，還有蝕雙星 (eclipsing binary, ECL)。和老師討論後，我們認為這種變星的統計意義較小，因此確認以分析 RRL 成員星比例 (membership ratio) 為研究主軸。 老師在這段期間有和我提到分析 RRL 可能可以推論出和恆星演化 (star formation) 有關的結論。當時我還不知道其關聯性，是到十月中得到結果後才比較有概念。 I found that in addition to RRL, most star clusters also have eclipsing binary (ECL) stars. After discussing with the teacher, we believed that the statistical significance of this type of variable star was small, so we confirmed that the analysis of the membership ratio of RRL stars would be the main focus of our research. During this period, the teacher mentioned to me that analyzing RRL might be able to deduce conclusions related to star formation. I didn't know the correlation at the time. I had a better idea only after I got the results in mid-October.	老師 幾乎可以說應該與星星數成正相關 9 月 23, 2023 9:36:55下午 老師 有一點，但不多 9 月 23, 2023 9:36:34下午 學生 所以是不是說不太具有統計意義呢？ 9 月 23, 2023 9:36:23下午 老師 就是雙星，但繞行盤面與視線方向平行 9 月 23, 2023 9:33:42下午 老師 ECL 是機率 9 月 23, 2023 9:33:20下午 學生 RR 是唯二每個星團都有的變星 另一個是 ECL (eclipsing binary) 之前您說過不是我們討論的變星 9 月 23, 2023 9:33:04下午 老師 star formation 9 月 23, 2023 9:39:46下午 老師 恆星演化 9 月 23, 2023 9:39:38下午 學生 那請問除了這些變星的比例還有甚麼可以探討的嗎 像是從這個數據不知道老師有沒有看到其他可以分析的東西 9 月 23, 2023 9:39:10下午 學生 好的 我再查一下！ 9 月 23, 2023 9:44:14下午 老師 一時想不到了 9 月 23, 2023 9:43:11下午 老師 變星能持續多久 9 月 23, 2023 9:42:53下午 老師 變星為什麼這麼少 9 月 23, 2023 9:42:42下午 老師 那是演化中末期發生了啥事 9 月 23, 2023 9:42:34下午 老師 若要跟變星有關 9 月 23, 2023 9:42:19下午
-----------------------------------	--	--

2023/09/29

我得到初步的 RRppm 數據，並嘗試找出其和金屬豐度、質量、星團半質量半徑的關聯。其中，我最先發現 RRppm 和半徑的正相關。

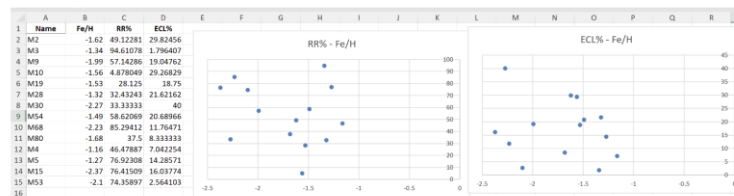
I got preliminary RRppm data and tried to figure out how it relates to metallicity, mass, and cluster half-mass radius. Among them, I first discovered the positive correlation between RRppm and radius.

學生

若我做的資料正確
單純看組成比例看不出明顯的關聯...

9 月 29, 2023 12:23:17 下午

學生



9 月 29, 2023 12:22:50 下午

老師

金屬豐度是否與變量的出現有關

9 月 29, 2023 12:20:14 下午

老師

我的意思是

9 月 29, 2023 12:20:01 下午

學生

所以可能這三個是同一件事情？

9 月 29, 2023 5:28:34 下午

老師

半徑 ... 這其實也許跟質量也有關？

9 月 29, 2023 5:27:27 下午

老師

質光關係

9 月 29, 2023 5:27:02 下午

老師

星團星等其實就是質量

9 月 29, 2023 5:26:57 下午

學生

下載檔案：
your_activity_across_facebook/messages/inbox/huangguanfu_10160632118289602/files/AgeComparisonGrantd_1032637544716736.xlsx

9 月 29, 2023 5:22:55 下午

學生

老師好，我發現
1. 原本預期會有關係的 Fe/H 看不出明顯趨勢
2. RR ppm 和星團半徑、星團星等、星團質量有關連性

9 月 29, 2023 5:20:55 下午

2023/09/29

此外，我提出了一個有趣的發現，即球狀星團半徑似乎趨向一最小值，不過這並非最終研究結果所強調的結論之一。

為量化半徑的正相關，老師建議我找出 power law 關係式。以 $\log RR_{ppm}$ 與 $\log$ 半徑 作圖，我發現的確有正相關。

In addition, I made an interesting finding that the radii of globular clusters seem to tend toward a minimum value, although this was not one of the conclusions emphasized by the final results.

In order to quantify the positive correlation of radius, the teacher suggested that I find the power law relationship. Plotting $\log RR_{ppm}$ versus $\log$ Radius, I found that there is indeed a positive correlation.

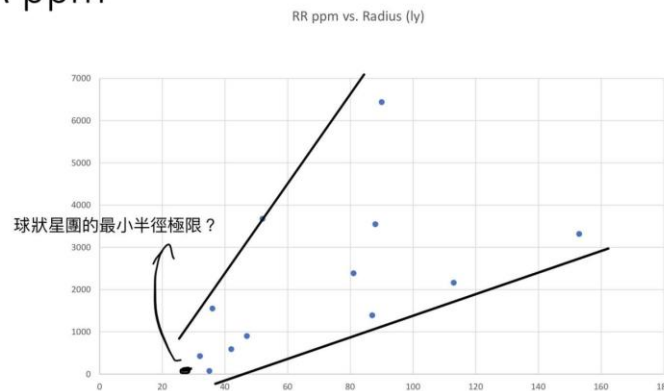
學生

這是一個我覺得有趣的地方

9月29, 2023 5:32:02下午

學生

RR ppm



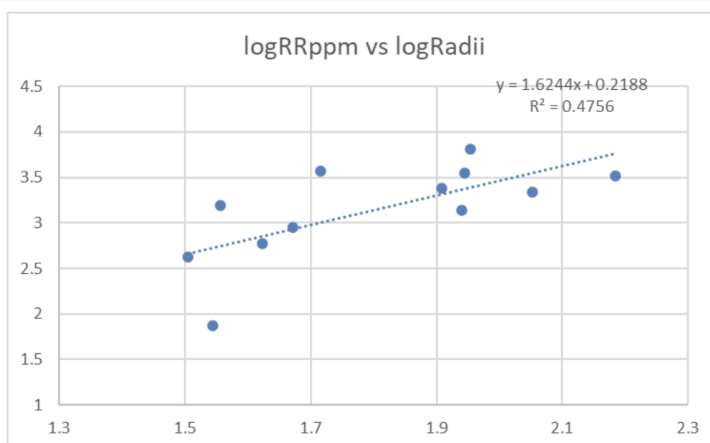
9月29, 2023 5:31:55下午

學生

好像真的有耶

9月29, 2023 5:45:35下午

學生



9月29, 2023 5:45:29下午

學生

我查到的是 $y = ax^k$ 的公式
如果 x, y 取 $\log$ 會預期有線性關係

9月29, 2023 5:45:24下午

老師

數量取個 $\log$

9月29, 2023 5:35:21下午

老師

不知，你先查一下 power law 的概念

9月29, 2023 5:35:11下午

2023/09/29

老師建議我同時考慮兩個參數，分析和 RRppm 的相關性，即在年齡的參數前乘上一參數。

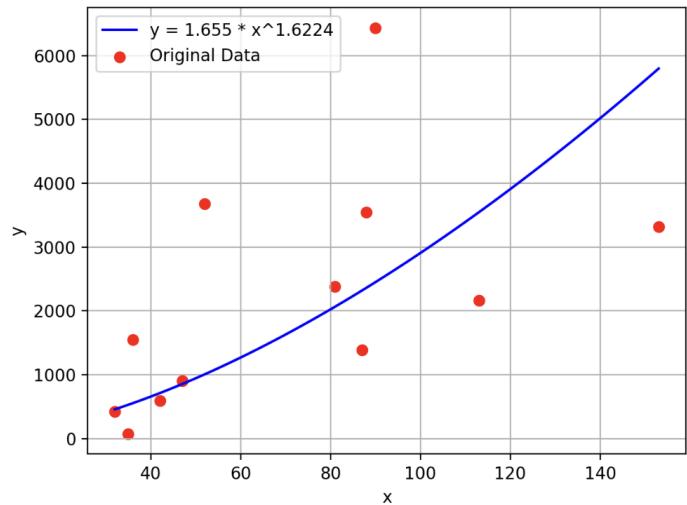
The teacher suggested that I consider two parameters at the same time and analyze the correlation with RRppm, that is, multiply the age parameter by a coefficient.

學生

所以我要把半徑和年齡用某個權重結合
然後一樣作圖
找相關係數最高的權重嗎？

9月29, 2023 6:40:07下午

學生



9月29, 2023 6:35:57下午

老師

或 1 0.7 0.3

9月29, 2023 6:31:07下午

老師

1 0.9 0.8

9月29, 2023 6:30:59下午

老師

或是取中間最大

9月29, 2023 6:30:51下午

老師

或反過來

9月29, 2023 6:30:45下午

老師

越年輕權重越小

9月29, 2023 6:30:42下午

老師

ex 130 或 140 億當1

9月29, 2023 6:30:32下午

老師

能不能加個年齡的權重項

9月29, 2023 6:30:18下午

老師

同時想一個理由為什麼會有關係

9月29, 2023 6:29:41下午

2023/09/29~
2023/10/15

此時我完成了大部份的「觀測」部分，也就是彙整了所有數據，Python 的係數 λ, μ 的優化 (optimization of coefficients) 部分也告一段落。

不過，我觀察數據趨勢的敏銳度較差，一開始以為 RRppm 只和年齡有負相關及和半徑有正相關。後來認真分析 X_{RR}, Y_{RR} 變數時，才發現 RRppm 與金屬豐度的關聯性。

At this point, I have completed most of the "observation" part, that is, aggregating all the data, and the optimization of coefficients λ and μ in Python has also come to an end.

However, my sensitivity in observing data trends is poor. At first, I thought RRppm was only negatively correlated with age and positively correlated with radius. Later, when I carefully analyzed the X_{RR} and Y_{RR} variables, I discovered the correlation between RRppm and metallicity.

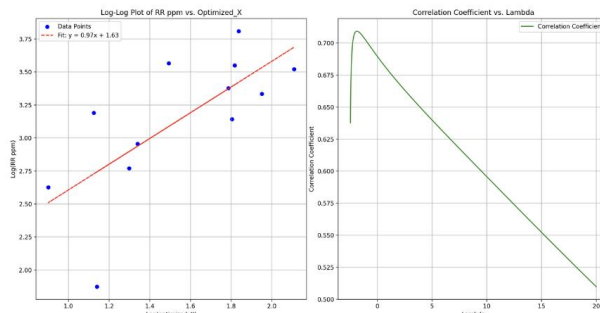
學生

老師好，我發現加了權重做雙變數的迴歸時，得到的最佳相關係數不會和只用半徑當變數時差太多
9月30, 2023 9:18:14上午

學生

RR ppm

$X = \text{半徑} + \lambda \times \text{年齡}$, 尋找使 $\log(\text{RRppm})$ vs. $\log(X)$ 相關係數最高的 λ
Optimized Lambda: -1.8589
Highest Correlation Coefficient: 0.7092
Regression: $y = 0.97x + 1.63$, 右圖在 λ 繼續變大時單調遞減



9月29, 2023 9:07:54下午

學生

似乎有一點 尤其是FeH 的圖有一點特別 不知道為什麼會長成U型
10月15, 2023 8:51:05上午

老師

看圖的趨勢

10月15, 2023 8:46:45上午

老師

我覺得其他也有負相關？

10月15, 2023 8:46:37上午

老師

這就是你的結論

10月15, 2023 8:45:33上午

學生

下載檔案：
your_activity_across_facebook/messages/inbox/huangguanfu_10160632118289602/files/starsummary1015_354826183637410.pdf

10月15, 2023 8:45:27上午

學生

老師好，我後來把星圖數增加到20個，目前數據都處理完了，但除了RRppm和年齡有一點負相關外，其他的似乎沒什麼明顯的趨勢，因此我不太確定要怎麼下結論.....

10月15, 2023 8:45:11上午

老師

我覺得你的觀測部份結束了

10月15, 2023 8:51:42上午

老師

但這需要理論來驗證

10月15, 2023 8:51:30上午

老師

也是個結論？

10月15, 2023 8:51:19上午

2023/10/15~ 2023/10/25	我整理資料，撰寫完實驗報告後，和老師討論是否有需要修改的地方。過程中，我和老師請教了有關引用 (citation)、撰寫動機與結論等。 After I organized the information and wrote the experimental report, I discussed with the teacher whether there were any changes that needed to be made. During the process, the teacher and I asked about citations, research motivation and conclusions, etc.	學生 review 的來源只有一篇嗎? 10 月 18, 2023 12:40:42下午 老師 來源是這篇 10 月 18, 2023 12:39:58下午 老師 那個是review 10 月 18, 2023 12:39:43下午 學生 vol 221 no 4 10 月 18, 2023 12:39:32下午 學生 好像不是 10 月 18, 2023 12:39:27下午 老師 https://articles.adsabs.harvard.edu/pdf/1996AJ....112.1487H %PDF-1.5 % 12 0 obj > endobj xref 12 13 0000000023 00000 n 0000000534 00000 n 0000000822 00000 n 0000000983 00000 n 0000001037 00000 n 0000001160 00000 n 0000002214 00000 n 0000002414 00000 n 0000002523 00000 n 0000009739 00000 n 0000012550 00000 n 0000017031 00000 n 0000000653 00000 n trailer https://articles.adsabs.harvard.edu/pdf/1996AJ....112.1487H 10 月 18, 2023 12:28:12下午 老師 其十分接近宇宙年 齡。因此，透過研究球狀星團，我們能夠對宇宙的演化窺知一二。 10 月 22, 2023 10:38:42下午 老師 分析其在“演化歷程”的研究 10 月 22, 2023 10:37:03下午 老師 上有重要意義 10 月 22, 2023 10:36:30下午 老師 摘要 且在“恆星演化” 10 月 22, 2023 10:36:23下午 老師 收到 10 月 22, 2023 10:29:08上午 學生 老師好，我目前只剩參考資料的部分，想請老師看一下摘要與結論的部分是否需要修改，謝謝老師 10 月 22, 2023 8:51:53上午 學生 下載檔案： your_activity_across_facebook/messages/inbox/huangguanfu_10160632118289602/files/20231022xingtuanke_zhancaogao_3235823690050930.pdf 10 月 22, 2023 8:42:04上午 學生 另外這是我目前的進度，有找到一個 power law，目前還要解釋半徑的正相關 請老師先看一下 10 月 18, 2023 4:15:12下午
---------------------------------------	---	--

2023/10/26~ 2023/10/30	進行研究報告的最終修改，主要圍繞在怎麼寫結論和報名資料填寫等。 另外，我和老師也討論了許多較細節的部份，如呈現相關係數、RRppm 範圍等。 The final revision of the research report mainly focuses on how to write the conclusion and fill in the registration information. In addition, the teacher and I also discussed many more detailed parts, such as the presentation of correlation coefficients, RRppm range, etc.	學生 老師好，請問這樣改可以嗎? 10 月 26, 2023 6:15:45下午 學生 以上研究結果說明除了一般被認為主導星團演化的年齡與金屬豐度會影響 RRppm 外，星團半徑亦和 RRL 於星團中的組成有正相關。目前對此現象的研究較少，推測其原因可能為 - 星雲說解釋：一團星雲形成眾多小星團。星團形成時，較大者會因萬有引力和較小者融合，因此半徑較大的星團成員星的組成較為多樣，可能有利於變星的形成。 - 偶發生成解釋：半徑較大的星團由於星體間萬有引力交互作用較顯著，成員星密度較高，使星體間碰撞、潮汐作用等交互作用較頻繁，進而導致變星的生成。不過，受限於觀測技術，在無法看到所有成員星的情況下，較難直接透過觀測成員星密度。 - 潮汐作用解釋：在半徑大的星團中，星體間潮汐作用較顯著，影響質量蒸發，使星團中較大比例的星體演化為變星。此假說呼應 Ferro 等人認為質量蒸發與 RRL 形成有關的論點。 以上三個假設中，潮汐作用解釋有較完備的理論背景，推測最為可能的機制。對於為何星團半徑與變星組成比例有正相關十分值得探討。 10 月 26, 2023 6:15:38下午 學生 好的 我修完再跟老師討論 10 月 26, 2023 4:16:30下午 學生 有 10 月 28, 2023 1:14:10下午 學生 有 但有意義嗎? 10 月 28, 2023 1:14:07下午 老師 真相關的相關係數? 10 月 28, 2023 1:13:55下午 老師 平均值有嗎 10 月 28, 2023 1:13:23下午 老師 變星的rpm 的範圍寫上去 10 月 28, 2023 1:13:05下午 學生 沒有 10 月 28, 2023 1:12:44下午 老師 偏差平均值有嗎 10 月 28, 2023 1:12:33下午 學生 老師請問結論這樣寫可以嗎? 10 月 28, 2023 1:11:50下午
-----------------------------------	--	---

2023/12/14~ 2023/12/30	通過初審後，製作英文簡報、英文解說影片等。 其中，我和老師在標題上討論許久，因為除了要強調「RR 型天琴座變星的成員星組成比例」外，也要讓標題簡潔有力，我還使用 ChatGPT 協助。 After passing the preliminary review of 2024 Taiwan Internatioal Science Fair, English briefings, English explanation videos, etc. were being produced. Among them, the teacher and I discussed the title for a long time, because in addition to emphasizing the "proportion of member stars of the RR Lyrae variable star", we also need to make the title concise and powerful. I also used ChatGPT to assist.	學生  12 月 14, 2023 12:24:21下午 老師 國際科展有過初審 12 月 14, 2023 12:13:18下午 學生 Compositional Analysis of RR Lyrae Variables in Globular Clusters Using GAIA Data? 12 月 25, 2023 6:55:32下午 老師 或你修一下 12 月 25, 2023 6:54:44下午 老師 percentage of variable star in the cluster? 12 月 25, 2023 6:54:36下午 老師 怪怪的 12 月 25, 2023 6:54:04下午 學生 You Turn this paper title into a more professional one: "Analyzing the Composition of RR Lyrae Variables in Globular Clusters with GAIA Data" ChatGPT To enhance the professionalism of the paper title "Analyzing the Composition of RR Lyrae Variables in Globular Clusters with GAIA Data," you could consider the following revision: "Investigating the Stellar Populations of RR Lyrae Variables within Globular Clusters Using GAIA Observations" This title maintains the essence of the original while employing more formal and specific scientific language. 12 月 25, 2023 6:53:52下午
-----------------------------------	--	---

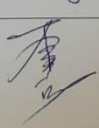
Main Remarks from Professor Li After TISF 2024 (2024/1 ~ 2024/10)

- Emphasize the discovery that the cluster radius is important.
- Change of symbols for key parameters.
- Presentation changes on the poster: table caption, figure caption.
- Analyze factor weights.
- Revised presentation script.

Notes

2024 年臺灣國際科學展覽會出國代表集訓輔導紀錄

時間：中華民國113年10月5日 ☒ 上午 ☐ 下午 (上下午分開寫) 地點：	
輔導教授： 李文献 教授	學生姓名： 何承祐
作品名稱：Star Cluster Evolution - Reviewed By GAIA	
科別：物理與天文學	就讀學校及年級：嘉義高中12年級
參加展覽會名稱：巴西科學博覽會 MOSTRATEC	
Note：請務必根據教授輔導內容修改，如未修正視同學生學習態度不良、效果欠佳者，經輔導教授團證明得取消其正選代表出國資格。(依臺灣國際科學展覽實施要點)	
請詳述輔導內容：	
標題：加上姓名，Cluster → Clusters	
Research Goals → Goals 加上 "Data from GAIA"	
Symbol & Physical Parameters 往前移	
去除 $\log R$ vs. Y 圖，使用 $\log R$ vs. X & $\log R$ vs. Z 強調 Radius	
不準確 有效位數取1位 X axis range 調成一格	
標示 yellow region = 95% confidence level	
$\log A = \log_{10} A$	
兩表格放在同一列	
Linear fit → fits	
Reference：放 Journal & Page number	
開頭："Good morning. ... research [TITLE]"	
symbols A age r radius M [Fe/H] R membership ratio	

教授簽名： 

Caption: Table 在 $\frac{r}{r}$, Fig 在下

Page 3

Linear Fit $\rightarrow$ Fits

不确定度只取到一位

Table 更小一点

2 Tables put on same line

$\Rightarrow$ 缩小垂直长度 Title 可能是同一张

删除 p-value column

可以把 Y 删除, 用 X 和 Z 来证明 radius 的效果

+ Key Physical Parameter

* Applying this method to consider the effect from radius to other star clusters.

拿掉 log 的 ()

Figure 6, (a) (c) X 轴 range 要一样 才可以比 半径 程度

1. Age

2. Meta.

3. Radius
框起来

Factor Weight

Factor/weight	Previous Result		This study	
	A	M	A	M
	1	2	NA	1

$\rightarrow$ in blue emph.

* Journal

Page

可以不写 Title

Background

RR Lyrae variable stars (RRLs), commonly found in globular clusters (GCs), are important stellar candles used for measuring cosmic distances. Globular clusters, which are dense, old, and metal-poor star clusters, are primarily located in the Milky Way's halo. Due to the limited astronomical data available in the past, the membership ratios of variable stars have not been fully investigated. This motivates the search for the ratio of RR Lyrae stars in globular clusters using the newly released GAIA database.

Research Goals

- Using GAIA Data Release 3, analyze the **membership ratio** r of RRLs to member stars in GCs.
- Determine which cluster parameters influence r most.
- Propose **hypotheses** regarding the cause of such result.

Methodology

Estimating Number of Stars in Globular Clusters

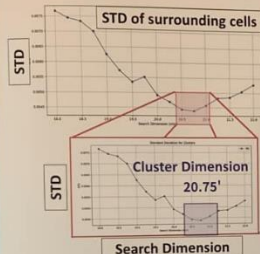


Figure 1 Estimating cluster dimension by calculating the dimension yielding lowest STD of stars surrounding the cluster.

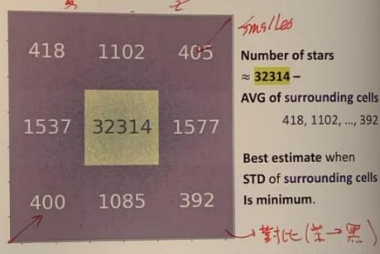


Figure 2 Heatmap of number of stars. Each number represents the number of stars in the 3 by 3 grid. Search dimension expanded until STD of surrounding cells reach minimum.

Isochrone Fitting

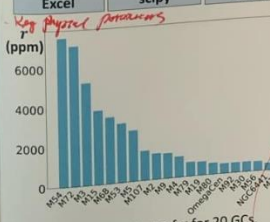


Figure 4 Bar chart of r for 20 GCs.

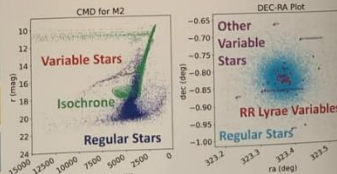


Figure 3 CMD (left) and DEC-RA plot (right) of cluster M2. The isochrone line was shifted to fit the blue dots. Variable stars that deviate from the isochrone line were removed.

Membership Ratio Analysis

$$X = a_X \cdot \log A + b_X \cdot F$$

$$Y = a_Y \cdot \log A + b_Y \cdot \log R$$

$$Z = a_Z \cdot \log A + b_Z \cdot F + c_Z \cdot \log R$$

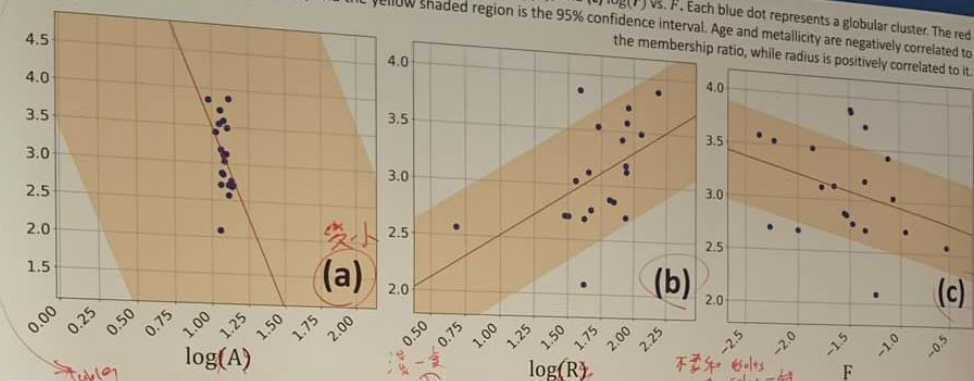
$$r = \frac{\text{Number of RRLs in GC}}{\text{Total Number of Stars in GC}} \times 10^6 \text{ ppm}$$

AVG	Average
STD	Standard deviation
RRL	RR Lyrae
GC	Globular Cluster
CMD	Color-Magnitude Diagram
DEC	Declination
RA	Right Ascension

Symbol	Meaning	Unit
A	Cluster age	Gyr (10^9 years)
R	Cluster half-light radius	Light Years (ly)
X	Cluster metallicity [Fe/H]	dex
r	Membership ratio of RRLs	Parts per million (ppm)

Results

Figure 5 Plots of (a) $\log(r)$ vs. $\log(A)$, (b) $\log(r)$ vs. $\log(R)$, and (c) $\log(r)$ vs. F . Each blue dot represents a globular cluster. The red line is the linear regression, and the yellow shaded region is the 95% confidence interval. Age and metallicity are negatively correlated to the membership ratio, while radius is positively correlated to it.



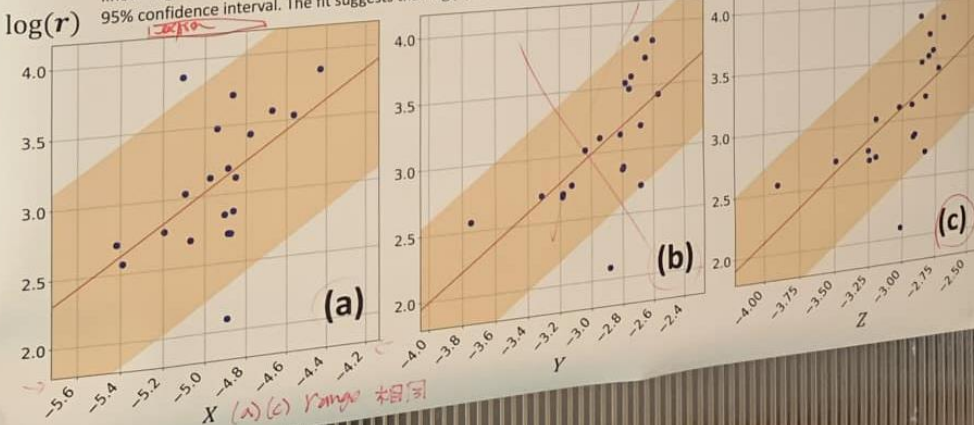
Quantity	a	b	c	Linear Fit	Correlation Coefficient	P-value
X	-6.264	-0.374	-	$\log(r) = (1.00 \pm 0.41)X + (8.03 \pm 2.01)$	0.501	0.0244
Y	-43.829	8.004	-	$\log(r) = (1.00 \pm 0.31)Y + (5.99 \pm 0.90)$	0.605	0.0047
Z	-71.096	11.110	-2.102	$\log(r) = (1.00 \pm 0.30)Z + (6.11 \pm 0.92)$	0.613	0.0040

Parameter	Value Range	Coefficient in Z	Relative Significance
$\log(A)$	0.978~1.152	-71.096	~1
$\log(R)$	0.681~2.185	11.110	~0.22
F	-2.37~0.95	-2.102	~0.05

Table 1 Optimal coefficients, linear fit, correlation coefficient and p-value between $\log(r)$ and X, Y, and Z. The small p-values indicate there is a statistically significant association between X, Y, Z and r.

Table 2 Relative significance of $\log(A)$, $\log(R)$ and F. Relative significance is estimated by the product of a quantity's coefficient in Z and its typical value range. Age, radius, and metallicity are primary factors that influence the membership ratio r, listed in decreasing order of impact.

Figure 6 Plots of (a) $\log(r)$ vs. X, (b) $\log(r)$ vs. Y, and (c) $\log(r)$ vs. Z. Each blue dot represents a globular cluster. The red line is the linear regression, and the yellow shaded region is the 95% confidence interval. The slope is normalized to 1. Most clusters fall inside the 95% confidence interval. The fit suggests that $\log(r)$ is moderately correlated with X, Y and Z.



Interpretation

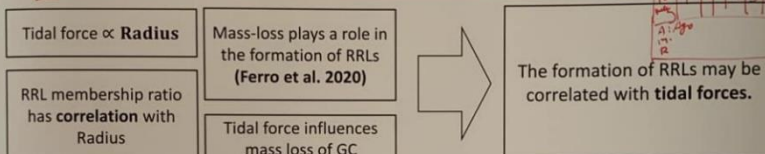
High r : M54 Cluster

- M54 has a high membership ratio with no significant age, radius or metallicity
- M54 belongs to the Sagittarius Dwarf Galaxy while others are in the Milky Way.
- The formation of RRLs may be influenced by its home galaxy.

[Fe/H] and the Instability Strip

- Negative correlation of number of RRLs with [Fe/H] can be explained with the **instability strip**:
- The instability strip is **narrower** when the metallicity is **higher**. [3]
 - Therefore **less variable stars** populate the instability strip, causing smaller r .

Correlation Between Radius and Membership Ratio



Low r : M10 Cluster

- M10 has a significantly small r . In agreement with past research:
- No RRLs in M10 found in 2002 [1] X
- Only one RRL found as of 2020 [2] X
- Hypothesis** X
 - It is claimed [2] that there may be variations on its Red Giant Branch (RGB) evolution.
 - M10 may have a unique mass loss due to, for example, collision with another cluster.

Cluster Parameters Influencing r

- Age**
Primary factor of GC evolution [4]
- Radius**
Tidal force & mass-loss
- Metallicity**
Secondary factor of GC evolution [4]

Conclusions

- Using **GAIA Data Release 3**, 20 globular clusters were analyzed.
- The number of stars in GCs were found with the method of **heatmaps**. After fitting with **isochrone curves**, the membership ratio r was calculated. By defining X, Y, and Z, the influence of cluster parameters was quantitatively analyzed.
- Age, Radius and Metallicity** were found to have the greatest influence on the RRL membership ratio, in decreasing order.
- The roles of **Age and Metallicity** is in agreement with literature.
- The impact of **Radius** can be explained with **mass-loss and tidal forces**.
- From the clusters that deviate from the main trend, it is concluded that **the home galaxy or RGB evolution** may influence RRL formation. Moreover, the **instability strip** explains how **metallicity** affects the RRL membership ratio.

Future Prospects

- Similar analyses could be conducted on open clusters, given that member stars can be properly distinguished from background and foreground stars.
- Apart from RRL, other types of variable stars could also be analyzed.
- Deeper analysis into globular clusters could reveal insights of the early ages of the universe.

References

- [1] Braun, K., et al. (2002). "Photometry Results for Globular Clusters M10 and M12."
- [2] Ferro, A. A., et al. (2020). "Globular Cluster M10: Reassessment of Stellar Membership, Distance, and Age."
- [3] Demarque, P., & Zinn, R. (1999). "Metallicity Dependence of RR Lyrae Absolute Magnitudes."
- [4] Gratton, R., et al. (2019). "Globular Cluster: An Observational Perspective."

This work has made use of data from the European Space Agency (ESA) mission Gaia (<https://www.cosmos.esa.int/gaia>), processed by the Gaia Data Processing and Analysis Consortium (DPAC, <https://www.cosmos.esa.int/web/gaia/dpac/consortium>). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement.

2024 年臺灣國際科學展覽會出國代表集訓輔導紀錄

時間：中華民國113年10月9日 ☐ 上午 ☒ 下午（上下午分開寫）		地點：科教館
輔導教授：	學生姓名：	
李文猷 教授	何承祐	
作品名稱：Star Cluster Evolution — Revealed by GAIA		
科別：	就讀學校及年級：	
物理與天文學科	嘉義高中 12 年級	
參加展覽會名稱：		
Note: 請務必根據教授輔導內容修改，如未修正視同學生學習態度不良、效果欠佳者，經輔導教授團證明得取消其正選代表出國資格。（依臺灣國際科展實施要點） 請詳述輔導內容： This work ... mission Gaia 框起來 $R = \dots \Rightarrow R \equiv \dots$ References 月份不用 no. 不用 pp 不用 Journal $\rightarrow$ J. This work ... agreement 分三句 References 重排 Factor Weight $\rightarrow$ Weight NA: Applicable This Research $\rightarrow$ This Work Fig 5 字 blue $\rightarrow$ darkblue Using GAIA ... 3 移到後面 Conclusion 第 = 之前加入 "I found R is a key factor that must be considered in understanding star cluster evolution." Figure 5 縮小 Figure 6 (b)(c) 易經		

教授簽名：

